A Study on Model Predictive Control Architectures for Passenger Autonomous Vehicles

Phurithat Tangsripairoje

***Abstract*—This paper presents a unified model predictive control (MPC) framework for highway driving that integrates lane keeping, lane changing, adaptive cruise control (ACC), and obstacle avoidance within a single optimization structure. The method combines a dynamic bicycle model with lateral Magic Formula tire forces, an ACC module for lead-vehicle following, and a geometric corridor-generation algorithm that enforces road and obstacle feasibility. A piecewise-affine reference adaptation ensures smooth transitions between avoidance maneuvers and nominal lane tracking. The MPC problem is solved using a linear time-varying approximation with propulsion, braking, and steering-rate regularization. Simulations across three scenarios—lane following, lane change with ACC, and obstacle avoidance—demonstrate high tracking accuracy, stable yaw dynamics, smooth longitudinal control, and physically consistent tire-force behavior. Even in complex avoidance cases, the vehicle respects corridor bounds, friction limits, and safe headway, indicating that MPC coupled with geometric corridor planning forms a robust and extensible foundation for autonomous highway driving.**

Github: https://github.com/winzard10/HighwayMPC

I. Introduction

Commercial autonomous vehicles operate in complex, volatile, and highly dynamic environments. They must negotiate continuously changing traffic conditions, interact safely with human drivers, and maintain comfort and stability while performing highway maneuvers such as lane keeping, lane changes, obstacle avoidance, and adaptive cruise control. Although companies such as Waymo, Tesla, Cruise, and others have demonstrated impressive levels of autonomous capability, the details of their control architectures remain proprietary. Much of the technical foundation behind how these systems achieve safe, smooth, and reliable behavior in real-world conditions is therefore not publicly accessible.

This report presents a focused study on the fundamentals of longitudinal and lateral motion control for autonomous vehicles using Model Predictive Control (MPC). The goal is not to replicate any particular commercial system, but to understand the essential components that form the backbone of modern self-driving car controllers. By constructing a complete simulation framework from first principles—including a dynamic bicycle model with tire forces, a highway environment with lane geometry, an obstacle-aware corridor planner, and an LTV–MPC controller—we gain insight into the structure, assumptions, and challenges inherent in real-world autonomous driving control systems.

The study investigates how MPC can integrate road geometry, tire physics, actuator constraints, comfort considerations, and safety requirements into a unified optimization-based controller. This offers a transparent and interpretable alternative to industry black boxes, enabling a deeper understanding of:

- how vehicle states evolve under realistic tire and steering dynamics,
- how predictive control manages trade-offs between tracking accuracy, stability, and comfort, and
- how obstacle avoidance and adaptive cruise control can be incorporated into the MPC formulation

Ultimately, this work provides a hands-on exploration of the control-theoretic principles that underpin highway-capable autonomous vehicles. By building, testing, and analyzing each module in a transparent simulation environment, we seek to bridge the gap between academic motion-planning literature and the high-level behavior observed in commercial autonomous systems.

II. Methodology

A. Highway Environment

In the U.S., AASHTO recommends a standard lane width of 12 ft (3.6 m) for U.S. freeways and multi-lane highways [1] while interstate and principal arterial freeways use a standard lane width of 12 ft with typical design speeds of 70-80 mph (113–129 km/h) [2]. For this study, we simulated a standard two-lane highway with a symmetric layout, a posted speed limit of $v_{ref} = 33$ m/s (approximately 120 km/h), and a lane width of $w_{lane} = 3.7$ m. The reference road centerline is constructed using a sinusoidal geometry to emulate long-range curvature variations that are typical in real-world freeways. The road shape is defined as

$$y_{\text{center}}(s) = A \sin\left(\frac{2\pi}{\lambda} s\right), \tag{1}$$

where $A = 1.5$ m is the lateral amplitude, $\lambda = 800$ m is the wavelength, and s is the road arclength. Right and left lane centerlines are generated by shifting the road centerline by ± 1.85 m.

Curvature $\kappa(s)$ and heading $\psi(s)$ are computed analytically and stored in a lane-center CSV file that is used by both the simulation and the trajectory planner. The Frenet (s, e_y) coordinate system is employed for control design.

B. Vehicle Model

TABLE I: Vehicle parameters and actuator limits.

Parameter	**Symbol**	**Value**
Sprung mass	m_s	660 kg
Wheelbase	L	3.4 m
CG distance from rear axle	d	1.6 m
Yaw inertia about CG	J_G	2500 kg m^2
Track width	w_{track}	1.7 m
Integration step	Δt	0.05 s
Max steering angle	$\delta_{\max}$	0.5 rad
Max steering rate	$\dot{\delta}_{\max}$	0.7 rad/s
Min longitudinal force	$R_{\min}$	-5000 N
Max longitudinal force	$R_{\max}$	5000 N
Front-left wheel force limit	$F_{\text{fl,max}}$	5000 N
Rear-left wheel force limit	$F_{\text{rl,max}}$	5500 N

Similar to the Frenet-frame modeling used in [3] and the state vector used in [4], the proposed framework focuses on the lateral slip dynamics while neglecting longitudinal tire-force coupling where its vehicle parameters are shown in Table I. However, whereas [3] relies on a linear cornering-stiffness approximation for small slip angles, we extend the model by integrating the full static Magic-Formula lateral tire force adopted in [5], where the central quantities of interest are the lateral deviation e_y and heading error e_ψ relative to the lane centerline as shown in Fig. 1. This modification improves the accuracy of the predicted lateral dynamics in scenarios where slip angles and load variations exceed the range of validity of the linear approximation.

a) State Vector.: The state vector is represented as

$$x = \begin{bmatrix} e_y \\ e_\psi \\ v_x \\ v_y \\ \dot{\psi} \\ \delta \\ d \end{bmatrix} \tag{2}$$

where

- e_y : lateral offset from the reference centerline,
- e_ψ : heading error,
- v_x : longitudinal velocity,
- v_y : lateral velocity,
- $\dot{\psi}$: yaw rate,
- δ : steering angle,
- d : distance gap to the lead vehicle (ACC only).

b) Input Vector.: The control inputs are

$$u = \begin{bmatrix} R \\ \dot{\delta} \end{bmatrix}, \tag{3}$$

where R is the longitudinal force command (acceleration or braking) and $\dot{\delta}$ is the commanded steering rate.

c) Nonlinear Dynamics.: Kinematic relationships in Frenet coordinates:

$$\dot{e}_y = v_x \sin(e_\psi) + v_y \cos(e_\psi), \tag{4}$$

$$\dot{e}_\psi = \dot{\psi} - \kappa_{\text{ref}} v_x. \tag{5}$$

Dynamic bicycle model equations:

$$\dot{v}_x = \frac{F_{x,f} + F_{x,r}}{m_{\text{total}}} + \dot{\psi}\, v_y, \tag{6}$$

$$\dot{v}_y = \frac{F_{y,f} + F_{y,r}}{m_{\text{total}}} - \dot{\psi}\, v_x, \tag{7}$$

$$\ddot{\psi} = \frac{M_z}{J_z}, \tag{8}$$

$$\dot{\delta} = \dot{\delta}_{\text{cmd}}, \tag{9}$$

ACC spacing dynamics:

$$\dot{d} = v_{\text{lead}} - v_x. \tag{10}$$

where

$$m_{total} = m_s + m_{us,f} + m_{us,r} \tag{11}$$

d) Longitudinal Load Transfer.: Normal forces at the front and rear axles are obtained in two stages. First, static axle loads $F^0_{z,f}$ and $F^0_{z,r}$ are computed from the sprung mass, gravity, wheelbase, center-of-gravity (CG)

position, and unsprung masses at each axle. Let m_s be the sprung mass, g the gravitational constant, L the wheelbase, d the distance from the rear axle to the CG, and $m_{\text{us},f}$, $m_{\text{us},r}$ the unsprung masses (per axle) at the front and rear, respectively. The static normal loads are

$$F^0_{z,f} = m_s g \, \frac{L-d}{L} + m_{\text{us},f} g, \tag{12}$$

$$F^0_{z,r} = m_s g \, \frac{d}{L} + m_{\text{us},r} g. \tag{13}$$

Then, longitudinal load transfer due to acceleration or braking is modeled using a simple linear relation with a tunable coefficient K_{zx} [5]. Given an approximate body-frame longitudinal acceleration a_x, the dynamic normal loads are

$$F_{z,f} = F^0_{z,f} - K_{zx} a_x, \tag{14}$$

$$F_{z,r} = F^0_{z,r} + K_{zx} a_x. \tag{15}$$

These dynamic normal loads are then used to scale the Magic Formula lateral force capacity at each axle and to enforce a friction ellipse constraint. The resulting longitudinal and lateral forces are rotated into the body frame and summed to obtain the total force (F_x, F_y) and yaw moment M_z.

For simplicity, the lateral load transfer is neglected.

TABLE II: Tire parameters and Magic Formula coefficients.

Parameter	**Symbol**	**Value**
Front friction coefficient	μ_f	0.9
Rear friction coefficient	μ_r	0.9
Magic Formula: front	(B_f, C_f, E_f)	$(13.0,\ 1.285,\ -0.9)$
Magic Formula: rear	(B_r, C_r, E_r)	$(13.0,\ 1.285,\ -0.9)$
Unsprung mass (front axle)	$m_{\text{us},f}$	120 kg
Unsprung mass (rear axle)	$m_{\text{us},r}$	80 kg
Load-transfer coefficient	K_{zx}	130.0

e) Tire Model.: Slip angles:

$$\alpha_f = \arctan\left(\frac{v_y + L_f \dot{\psi}}{v_x}\right) - \delta, \tag{16}$$

$$\alpha_r = \arctan\left(\frac{v_y - L_r \dot{\psi}}{v_x}\right). \tag{17}$$

Magic Formula lateral forces:

$$F_{y,i} = -2D_i \sin\Big(C_i \arctan\Big(B_i \alpha_i - E_i\big(B_i\alpha_i - \arctan(B_i\alpha_i)\big)\Big)\Big), \qquad i \in \{f, r\}. \tag{18}$$

where B_i, C_i, D_i, and E_i from Table II are the Magic Formula shape parameters for the front ($i = f$) and rear ($i = r$) axles [6]. In the implementation, slip angles are computed from the body-frame velocities (v_x, v_y) and yaw rate $\dot{\psi}$ using a dynamic bicycle geometry with front and rear axle distances $L_f = L - d$ and $L_r = d$ from the center of gravity (CG).

The peak term D_i is chosen proportional to the available normal load at each axle and the corresponding friction coefficient. Let μ_f and μ_r denote the front and rear friction coefficients, and $F_{z,f}$, $F_{z,r}$ the dynamic normal loads obtained from the load-transfer model. Assuming two wheels per axle, the per-axle peak scales are implemented as

$$D_f = \mu_f \, \frac{F_{z,f}}{2}, \qquad D_r = \mu_r \, \frac{F_{z,r}}{2}, \tag{19}$$

and the Magic Formula is evaluated per wheel and then multiplied by 2 to recover the total lateral force per axle, as reflected in (18) by the factor of $-2D_i$.

In this work, we assume a rear-wheel-drive (RWD) architecture for longitudinal force generation. During acceleration, all meaningful tractive effort is applied at the rear axle [7]:

$$F_{x,f} \approx 0, \qquad F_{x,r} = R_{\text{cmd}}, \qquad R_{\text{cmd}} > 0. \tag{20}$$

Although freely rolling front wheels do require small longitudinal friction forces to spin up and overcome rolling resistance, these forces are negligible compared to propulsion forces and are therefore omitted in the model. The front axle thus contributes only lateral force in the dynamic bicycle representation.

During braking ($R_{\text{cmd}} < 0$), a more realistic front-biased distribution is employed to reflect hydraulic brake proportioning and the increase in front normal load under deceleration. A 60:40 split is used [8]:

$$F_{x,f} = 0.6\, R_{\text{cmd}}, \qquad F_{x,r} = 0.4\, R_{\text{cmd}}, \qquad R_{\text{cmd}} < 0. \tag{21}$$

Longitudinal tire slip is not modeled explicitly. In high-speed highway maneuvers, the slip ratio required to generate moderate acceleration (1–3 m/s^2) is extremely small (typically below 5%) [6], [9], [10], and its influence on lateral behavior is negligible compared to the

effect of slip angle. As a consequence, longitudinal force is treated as a direct control input while only lateral slip is passed through the Magic Formula. This simplification is widely used in lateral control and MPC-based highway models, where the dominant nonlinearities arise from cornering forces rather than from tractive-slip dynamics.

In addition to pure lateral behavior, combined longitudinal–lateral effects are modeled via a friction ellipse constraint at each axle [11],

$$\sqrt{F_{x,i}^2 + F_{y,i}^2} \;\le\; \mu_i\, F_{z,i}, \qquad i \in \{f, r\}, \tag{22}$$

where $F_{x,i}$ and $F_{y,i}$ denote the longitudinal and lateral forces in the tire frame, $F_{z,i}$ is the normal load, and μ_i the corresponding friction coefficient. Given a requested pair $(F_{x,i}^0, F_{y,i}^0)$ from longitudinal control and the Magic Formula lateral model, we project the force back onto the friction ellipse via a scalar factor

$$\lambda_i = \min\left\{1,\ \frac{\mu_i F_{z,i}}{\sqrt{(F_{x,i}^0)^2 + (F_{y,i}^0)^2}}\right\}, \tag{23}$$

$$[F_{x,i}, F_{y,i}]^\top = \lambda_i [F_{x,i}^0, F_{y,i}^0]^\top. \tag{24}$$

For the front axle, these saturated forces are computed in the tire frame and then rotated into the vehicle body frame using the steering angle δ,

$$F_{x,f}^{\text{body}} = F_{x,f}^{\text{tire}} \cos\delta - F_{y,f}^{\text{tire}} \sin\delta, \tag{25}$$

$$F_{y,f}^{\text{body}} = F_{x,f}^{\text{tire}} \sin\delta + F_{y,f}^{\text{tire}} \cos\delta. \tag{26}$$

The rear axle is aligned with the vehicle body frame, so that

$$F_{x,r}^{\text{body}} = F_{x,r}^{\text{tire}}, \tag{27}$$

$$F_{y,r}^{\text{body}} = F_{y,r}^{\text{tire}}. \tag{28}$$

Summing front and rear contributions gives the total longitudinal and lateral forces and the yaw moment about the center of gravity (CG),

$$F_x = F_{x,f}^{\text{body}} + F_{x,r}^{\text{body}}, \tag{29}$$

$$F_y = F_{y,f}^{\text{body}} + F_{y,r}^{\text{body}}, \tag{30}$$

$$M_z = L_f\, F_{y,f}^{\text{body}} - L_r\, F_{y,r}^{\text{body}}, \tag{31}$$

where $L_f = L - d$ and $L_r = d$ are the distances from the CG to the front and rear axles, respectively. These quantities (F_x, F_y, M_z) are then used as inputs to the dynamic bicycle model.

C. Lane-Change Reference Generation

To enable smooth and kinematically consistent lane changes, the planner generates a time-parameterized reference trajectory that blends between the *from-lane* and the *to-lane* centerlines. Let t_{change} denote the nominal lane-change start time and T_{change} the desired transition duration. For $t < t_{\text{change}}$ the reference follows the original lane, while for $t > t_{\text{change}} + T_{\text{change}}$ the reference coincides with the target lane.

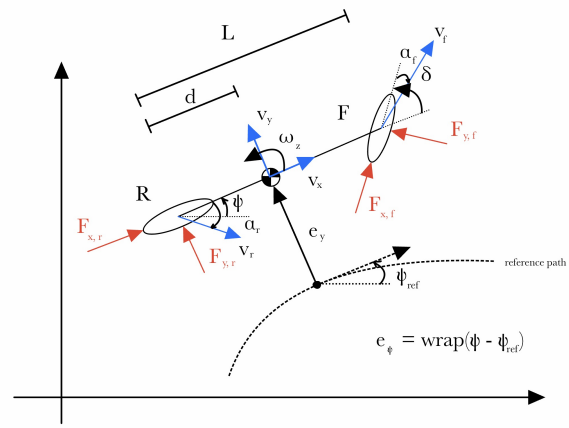


Fig. 1: Dynamic bicycle model with lateral tire forces model in Frenet coordinate.

Empirical studies of naturalistic driving show that typical lane-change durations are on the order of 5–6 s (e.g., mean 6.3 s). Therefore, designing the trajectory generation to complete a lane change in 5 s is consistent with observed human driving [12].

During the transition interval, a smooth interpolation is used.

a) Blend factor.: Define the normalized transition progress

$$\alpha(t) = \text{smoothstep}\left(\frac{t - t_{\text{change}}}{T_{\text{change}}}\right) \in [0, 1], \tag{32}$$

where $\text{smoothstep}(\cdot)$ is a C^1-continuous polynomial ramp ensuring zero slope at the start and end of the maneuver. This avoids abrupt changes in the reference and reduces transients in the closed-loop controller.

b) Lane centerline poses.: At each sampling time the vehicle's current position (x, y) is projected onto the map's Frenet coordinate system to obtain the corresponding centerline station s. Using the same station, the planner queries both the origin lane and the target lane to obtain their geometric poses,

$$(x_{\text{from}}(s), y_{\text{from}}(s), \psi_{\text{from}}(s)), (x_{\text{to}}(s), y_{\text{to}}(s), \psi_{\text{to}}(s)),$$

where ψ denotes heading.

c) Geometric blending.: The reference trajectory is obtained by convex combination of the two lane poses. Position is blended linearly,

$$x_{\text{ref}} = (1 - \alpha) x_{\text{from}} + \alpha x_{\text{to}}, \tag{33}$$

$$y_{\text{ref}} = (1 - \alpha) y_{\text{from}} + \alpha y_{\text{to}}, \tag{34}$$

while heading is blended using the shortest-angle interpolation,

$$\psi_{\text{ref}} = \psi_{\text{from}} + \alpha\, \text{wrap}\big(\psi_{\text{to}} - \psi_{\text{from}}\big), \tag{35}$$

where $\mathrm{wrap}(\cdot)$ returns the equivalent angle in $(-\pi, \pi]$. This ensures continuity of orientation even when the lane headings cross $\pm\pi$.

d) Error computation.: Given the reference pose $(x_{\mathrm{ref}}, y_{\mathrm{ref}}, \psi_{\mathrm{ref}})$, the lateral and heading errors used by the controller are defined in Frenet form,

$$e_y = (x - x_{\mathrm{ref}})(-\sin\psi_{\mathrm{ref}}) + (y - y_{\mathrm{ref}})(\cos\psi_{\mathrm{ref}}), \quad (36)$$

$$e_\psi = \mathrm{wrap}(\psi - \psi_{\mathrm{ref}}). \quad (37)$$

These errors drive the MPC during the transition, causing the vehicle to track a continuously deforming geometric reference that smoothly shifts from one lane to the other.

The proposed blending scheme provides a compact and effective mechanism for lane-change execution. It guarantees:

- continuous and smooth reference evolution,
- geometric consistency with the map-defined lanes,
- bounded curvature during the transition (due to C^1 smoothstep),
- stable error feedback, since e_y and e_ψ are computed with respect to an always-well-defined moving lane frame.

This approach is computationally inexpensive and integrates seamlessly with the MPC formulation, enabling reliable lane changes in real time.

D. Obstacle avoidance

Obstacle avoidance is handled by a two-stage *corridor planning* module that converts a set of obstacles into lateral constraints in the Frenet frame and then selects a smooth, collision-free reference path $e_{y,\mathrm{ref}}(s)$ inside this corridor.

TABLE III: Corridor planning parameters.

Name	Value	Description
`margin`	0.10 m	Safety margin added to inflated obstacles.
L_{look}	70.0 m	Look-ahead distance for obstacle inclusion in the corridor.
`slope`	0.05 m/step	Maximum temporal variation allowed in $[\ell_k, u_k]$.

a) Obstacle representation.: Obstacles are loaded from a CSV file with fields `x0`, `y0`, `radius`. The field `radius` denotes a true radius that includes the physical size of the obstacle. At any time t, the obstacle module enumerates all obstacles that are active in this time window and returns their current positions and radii.

b) Projection to Frenet space.: For each MPC step k in the prediction horizon, the planner has a reference centerline point $p_{\mathrm{ref}}(s_k) = (x_{\mathrm{ref},k}, y_{\mathrm{ref},k})$ and heading $\psi_{\mathrm{ref},k}$. All active obstacles at the corresponding time t_k are projected into the local Frenet frame defined by $(p_{\mathrm{ref}}(s_k), \psi_{\mathrm{ref},k})$:

- the along-track coordinate ds is obtained by projecting the vector from $p_{\mathrm{ref}}(s_k)$ to the obstacle position onto the tangent direction;
- the lateral coordinate $e_{y,\mathrm{obs}}$ is obtained by projecting the same vector onto the normal direction.

Only obstacles within a finite look-ahead window $ds \in [-L_{\mathrm{look}}/3, L_{\mathrm{look}}]$ are considered relevant for the corridor at step k. In this simulation. The lateral coordinates $e_{y,\mathrm{obs}}$ are stored and reused by the corridor builder.

The obstacle-aware lateral corridor is parameterized by the values in Table III, where `slope` controls the maximum per-step change allowed in the lateral bounds during temporal smoothing. A smaller value enforces smoother variation of the bounds ℓ_k and u_k, while larger values allow sharper lateral transitions.

c) Lane-aware corridor bounds.: Before accounting for obstacles, the controller defines nominal lateral bounds $[\ell_k, u_k]$ at each step k based on lane geometry and the chosen lane reference (left/right). These encode which region of the road is allowed for the ego vehicle in pure lane-keeping. The function `buildRawBounds` then *tightens* these bounds using the obstacle disk model. Each obstacle is treated as a disk of radius

$$r_{\mathrm{eff}} = r_{\mathrm{obs}} + \frac{w_{\mathrm{track}}}{2} + \mathtt{margin}, \quad (38)$$

where r_{obs} is the obstacle radius, w_{track} is the vehicle track width, and `margin` is used for an additional safety inflation.

Using the sign and magnitude of $e_{y,\mathrm{obs}}$, the corridor builder determines whether the obstacle lies to the left or right of the lane centerline and shrinks ℓ_k or u_k accordingly, while respecting lane boundaries and distinguishing between left, right, and center lanes. This produces raw bounds $[\ell_k^{\mathrm{raw}}, u_k^{\mathrm{raw}}]$ that encode both lane limits and obstacle avoidance constraints. A temporal smoothing step (`smoothBounds`) then applies a slope limit on ℓ_k and u_k along the horizon so that the corridor does not change abruptly from one step to the next.

d) Graph-based corridor centerline.: The layered-graph construction and heading-based cost follow the geometric corridor planning approach introduced in [13] as Algorithm 1. Given the smoothed bounds $[\ell_k, u_k]$ at each step k, a graph-based planner (`planGraph`) chooses a smooth lateral reference $e_{y,\mathrm{ref}}(s_k)$ inside the corridor. At each step k, three candidate lateral positions are defined as

$$e_y^{(0)}(k) = \ell_k, \quad e_y^{(1)}(k) = \tfrac{1}{2}(\ell_k + u_k), \quad e_y^{(2)}(k) = u_k, \quad (39)$$

representing the lower edge, middle, and upper edge of the corridor. These candidates form a layered graph over the horizon, where each node corresponds to a

pair (k, c) (time index and candidate index). Directed edges connect nodes (k, c) to (k', c') with $k' > k$ if the straight segment between $e_y^{(c)}(k)$ and $e_y^{(c')}(k')$ stays entirely inside the bounds $[\ell_m, u_m]$ for all intermediate steps $m \in [k, k']$. The edge cost penalizes changes in heading, approximated by the angle of the segment in (s, e_y)-space. A dynamic programming (DP) pass finds the minimum-cost path through this acyclic graph, and backtracking yields a globally consistent, piecewise-smooth lateral reference $e_{y,\text{ref}}(s_k)$ that

- remains inside the obstacle-aware corridor at all steps, and
- minimizes unnecessary curvature or weaving.

Algorithm 1 `planGraph`: Graph-based corridor centerline planning

Require: Centerline grid $\{s_k\}$, raw bounds $\ell_k^{\text{raw}}, u_k^{\text{raw}}$
Ensure: Smoothed bounds ℓ_k, u_k and reference $e_{y,\text{ref}}(s_k)$

$(\ell_k, u_k) \leftarrow$ `smoothBounds`$(\ell_k^{\text{raw}}, u_k^{\text{raw}}, \texttt{slope})$
for $k = 0$ to $N - 1$ **do**
  $e_y^{(0)}(k) \leftarrow \ell_k$, $e_y^{(1)}(k) \leftarrow \frac{1}{2}(\ell_k + u_k)$, $e_y^{(2)}(k) \leftarrow u_k$
Initialize bestCost$[k, c] \leftarrow \infty$, pred$[k, c] \leftarrow$ `null`
bestCost$[0, c] \leftarrow 0$ for $c \in \{0, 1, 2\}$
for $k = 0$ to $N - 2$ **do**
  for $c = 0$ to 2 **do**
    if bestCost$[k, c] = \infty$ **then continue**
    for $k' = k + 1$ to $N - 1$ **do**
      $\Delta s \leftarrow s_{k'} - s_k$
      for $c' = 0$ to 2 **do**
        if not `segmentInside`$(\ell, u, k, k', e_y^{(c)}(k), e_y^{(c')}(k'))$ **then continue**
        $\Delta\theta \leftarrow \arctan\big((e_y^{(c')}(k') - e_y^{(c)}(k))/\Delta s\big)$
        $J \leftarrow$ bestCost$[k, c] + |\Delta\theta|$
        if $J <$ bestCost$[k', c']$ **then**
          bestCost$[k', c'] \leftarrow J$
          pred$[k', c'] \leftarrow (k, c)$
Backtrack from minimal-cost terminal node to obtain $e_{y,\text{ref}}(s_k)$
return $\{\ell_k, u_k, e_{y,\text{ref}}(s_k)\}$

e) Piecewise-affine reference adaptation.: The DP-derived path $e_{y,\text{road}}(s_j)$ ensures global corridor feasibility but may differ from the vehicle's current lateral error $\hat{e}_{y,k}$. Following [13], a piecewise-affine (PWA) adaptation blends the current error into the road reference over the horizon (`PWAref`), reducing abrupt reference jumps seen by the MPC, as shown in Algorithm 2.

f) Integration with MPC.: The final output of the corridor planner is a set of lateral bounds $[\ell_k, u_k]$ and a centerline recommendation $e_{y,\text{ref}}(s_k)$. These are passed to the LTV-MPC as:

1) hard inequality constraints $\ell_k \le e_y(k) \le u_k$ on the lateral error, and
2) a soft tracking reference $e_{y,\text{ref}}(s_k)$ in the stage cost.

As a result, the MPC is free to deviate from the original lane center to avoid obstacles, but is always confined to a collision-free corridor that respects the road geometry and lane structure.

Algorithm 2 `adaptEyRefPWA`: Piecewise-affine reference adaptation

Require: Road reference $\{e_{y,\text{road}}(j)\}_{j=0}^{N}$, current error $\hat{e}_{y,k}$
Ensure: Adapted reference $\{e_{y,\text{ref}}(j)\}_{j=0}^{N}$

if $N \le 3$ **then**
  for $j = 0$ to N **do**
    $\alpha \leftarrow j/N$
    $e_{y,\text{ref}}(j) \leftarrow (1 - \alpha)\hat{e}_{y,k} + \alpha e_{y,\text{road}}(j)$
  return
for $j = 0$ to N **do**
  $e_r \leftarrow e_{y,\text{road}}(j)$
  if $j \le 3$ **then**
    $\alpha \leftarrow j/3$
    $e_{y,\text{ref}}(j) \leftarrow \hat{e}_{y,k} + 0.5(e_r - \hat{e}_{y,k})\alpha$
  else
    $\alpha \leftarrow (j - 3)/(N - 3)$
    $e_{\text{mid}} \leftarrow 0.5(\hat{e}_{y,k} + e_r)$
    $e_{y,\text{ref}}(j) \leftarrow e_{\text{mid}} + 0.5(e_r - \hat{e}_{y,k})\alpha$
return $\{e_{y,\text{ref}}(j)\}$

E. Adaptive Cruise Control

Adaptive Cruise Control (ACC) is implemented as a lightweight module that provides a *lead-vehicle model* and a *gap state* to the LTV–MPC controller. When ACC is enabled, the vehicle not only tracks the road speed profile but also maintains a safe distance to a lead vehicle.

TABLE IV: Piecewise lead-vehicle speed profile.

Segment	Speed	Time Range
Segment 1	$v_1 = 33$ m/s	$t < 10$ s
Segment 2	$v_2 = 20$ m/s	$10 \le t < 20$ s
Segment 3	$v_3 = 28$ m/s	$20 \le t < 40$ s

a) Lead-vehicle model.: The lead vehicle is represented by a scalar speed profile $v_{\text{lead}}(t)$. A piecewise profile with up to three linear segments (v_1, v_2, v_3) over specified breakpoints (t_1, t_2, t_3), which allows scripted scenarios such as braking or accelerating traffic as shown in Table IV.

At any time t, a flag indicates whether a lead vehicle is present. During MPC preparation, this model is sampled over the prediction horizon $t_k = t_0 + k\Delta t$ to build a preview of the lead speed, $v_{\text{lead}}(t_k)$, and an indicator $\texttt{has_obj}[k] \in \{0, 1\}$ that tells the controller when ACC constraints should be active.

TABLE V: Adaptive Cruise Control parameters.

Parameter	Symbol	Value
Time headway	τ	1.4 s
Minimum distance	$d_{\min}$	5 m
Initial spacing	d_{init}	150 m

b) Gap dynamics.: ACC introduces a longitudinal spacing state $d(t)$ which represents the distance between the ego vehicle and the lead vehicle. Its continuous-time dynamics follow the standard relative-position relationship

$$\dot{d} = v_{\text{lead}} - v_x, \tag{40}$$

where v_{lead} is the lead speed and v_x is the ego longitudinal speed. In discrete time, the ACC module integrates this state using

$$d_{k+1} = d_k + \Delta t\,(v_{\text{lead},k} - v_{x,k}), \quad d_k \geq 0, \tag{41}$$

with d clamped to zero to avoid negative gaps. The internal gap state is initialized from the parameter d_{init} and then propagated forward as the simulation runs.

c) Time-gap policy and MPC coupling.: ACC parameters are collected in a struct $(\tau, d_{\min}, d_{\text{init}})$, where τ is a time headway and $d_{\min}$ is a standstill distance. When ACC is active, the desired gap is defined as

$$d_{\text{des}}(v_x) = d_{\min} + \tau\, v_x, \tag{42}$$

as the parameters are shown in Table V and the MPC cost includes a quadratic penalty on the gap error $d - d_{\text{des}}(v_x)$, together with a penalty on the relative speed $v_x - v_{\text{lead}}$ whenever a lead is present. This encourages the ego vehicle to adjust its propulsion command such that:

- the time headway τ is approximately maintained at cruising speeds,
- the spacing does not shrink below $d_{\min}$, and
- the ego does not aggressively close in on a slower lead vehicle.

In the absence of a lead vehicle, the ACC terms are effectively disabled and the controller reverts to pure speed-profile tracking on the highway.

A time headway of τ = 1.4 s is consistent with values used in both production ACC systems and academic research. Commercial ACC implementations from major OEMs typically use selectable headways between 1.0 and 2.0 s, and traffic-safety guidelines for human drivers (e.g., the widely taught "2-second rule") provide an upper bound for comfortable car-following. Human-factors studies have shown that headways in the range of 1.2–1.5 s offer a good balance between safety, comfort, and traffic throughput [14], [15], [16]. Headways below 1 s tend to cause aggressive accelerations and reduced stability margins, while headways above 2 s lead to overly conservative following and degraded highway capacity [17]. Therefore, selecting τ = 1.4 s places the controller squarely within accepted practice for ACC design and aligns with validated car-following behavior reported in the literature.

III. LTV–MPC Highway Controller

The highway controller is implemented as a receding-horizon Linear Time-Varying Model Predictive Controller (LTV–MPC) built on top of the nonlinear dynamic bicycle model described in the previous section. At each sampling instant, the controller:

1) builds a local linear approximation of the nonlinear dynamics along a reference trajectory over a finite horizon,
2) discretizes the linear model with zero-order hold (ZOH),
3) assembles a quadratic cost and a set of linear constraints capturing lane-keeping, comfort, and safety requirements, and
4) solves a sparse quadratic program (QP) using OSQP to obtain the optimal control sequence.

Only the first control input (longitudinal effort and steering rate) is applied to the vehicle, and the process is repeated at the next time step.

The MPC state is

$$x = \begin{bmatrix} e_y & e_\psi & v_x & v_y & \dot{\psi} & \delta & d \end{bmatrix}^\top, \tag{43}$$

where e_y and e_ψ are the lateral and heading errors in Frenet coordinates, (v_x, v_y) are the body-frame velocities, $\dot{\psi}$ is the yaw rate, δ is the steering angle, and d is the distance to a lead vehicle. The inputs are

$$u = \begin{bmatrix} R_{\text{cmd}} & \dot{\delta}_{\text{cmd}} \end{bmatrix}^\top, \tag{44}$$

where R_{cmd} is a "generalized force" command (driving or braking effort) and $\dot{\delta}_{\text{cmd}}$ is the steering rate command.

A nominal trajectory is constructed at each stage by assuming zero lateral error and zero sideslip, assigning the reference speed v_{ref}, and choosing a steering angle consistent with the road curvature ($\delta \approx L\,\kappa_{\text{ref}}$); this (x_0, u_0) pair serves as the linearization point for the LTV model as follows:

$$x_0 = \begin{bmatrix} 0 & 0 & v_{\text{ref}} & 0 & 0 & L\kappa_{\text{ref}} & 0 \end{bmatrix}^\top, \tag{45}$$

$$u_0 = \begin{bmatrix} 0 & 0 \end{bmatrix}^\top. \tag{46}$$

TABLE VI: MPC configuration parameters.

Parameter	Symbol	Value
Prediction horizon length	N	100
Sampling time	Δt	(inherits from vehicle)
Lateral error weight	w_y	2.0
Heading error weight	w_ψ	0.1
Speed tracking weight	w_v	2.5
Input magnitude (R)	w_R	1×10^{-4}
Input rate ($\dot{R}$)	$w_{\mathrm{d}R}$	1×10^{-5}
Input jerk ($\ddot{R}$)	$w_{\mathrm{dd}R}$	2×10^{-5}
Steering rate weight	w_{dd}	12.0
Steering rate change weight	w_{ddd}	5.0
Terminal e_y weight	$w_{y,f}$	3.0
Terminal e_ψ weight	$w_{\psi,f}$	8.0
ACC slack weight	w_{acc}	10.0
Speed limits	$v_{\min}, v_{\max}$	0, 40 m/s
Fallback lateral bound	$e_{y,\max}$	1.8 m

a) Cost function.: The LTV–MPC minimizes a quadratic tracking and regularization cost over a finite horizon of length N. Let $w_y, w_\psi, w_v, w_R, w_{dd}, w_{dR}, w_{ddd}, w_{ddR}, w_{\text{acc}}, w_{y,f}, w_{\psi,f}$ denote nonnegative weights. The stage and terminal cost

$$\begin{aligned} J \;=\; & \sum_{k=0}^{N-1} \Big(w_y \,(e_{y,k} - e_{y,k}^{\text{ref}})^2 \\ & + w_\psi \, e_{\psi,k}^2 + w_v \,(v_{x,k} - v_{x,k}^{\text{ref}})^2 \\ & + w_R \, R_k^2 + w_{dd} \, \dot{\delta}_k^2 + w_{\text{acc}} \, s_{\text{acc},\,k}^2 \Big) \\ & + \sum_{k=1}^{N-1} \Big(w_{dR} \,(R_k - R_{k-1})^2 + w_{ddd} \,(\dot{\delta}_k - \dot{\delta}_{k-1})^2 \Big) \\ & + \sum_{k=2}^{N-1} w_{ddR} \,\big(R_k - 2R_{k-1} + R_{k-2}\big)^2 \\ & + w_{y,f} \,(e_{y,N} - e_{y,N}^{\text{ref}})^2 + w_{\psi,f} \, e_{\psi,N}^2, \end{aligned} \tag{47}$$

where $s_{\text{acc},\,k}$ is the ACC slack variable. For additional longitudinal smoothness, a discrete "jerk" penalty on R_k is included via the second-order difference term in the third sum. All MPC parameters are shown in Table VI

b) Constraints.: The optimization is subject to linear dynamics and a collection of state/input constraints that encode lane-keeping, comfort, and safety, mostly adopted from [5].

Dynamics and initial condition: For each stage $k = 0, \ldots, N-1$, the ZOH-discretized linearized model is

$$x_{k+1} = A_k x_k + B_k u_k + c_k, \tag{48}$$

with initial condition fixed by the measured MPC state x_0,

$$x_0 = x_{\text{init}}. \tag{49}$$

Lateral corridor and lane bounds: The obstacle/corridor planner provides time-varying lateral bounds $e_{y,k}^{\min}$ and $e_{y,k}^{\max}$ that define a collision-free corridor in e_y for $k = 0, \ldots, N-1$:

$$e_{y,k}^{\min} \;\le\; e_{y,k} \;\le\; e_{y,k}^{\max}. \tag{50}$$

When no corridor is available, a symmetric lane-width bound $|e_{y,k}| \le e_y^{\max}$ is used instead.

Input and state box constraints: The longitudinal effort R_k and steering rate $\dot{\delta}_k$ are bounded at each stage,

$$R_{\min} \;\le\; R_k \;\le\; R_{\max}, \tag{51}$$

$$-\dot{\delta}_{\max} \;\le\; \dot{\delta}_k \;\le\; \dot{\delta}_{\max}, \tag{52}$$

and the steering angle is limited for all $k = 0, \ldots, N$:

$$-\,\delta_{\max} \;\le\; \delta_k \;\le\; \delta_{\max}. \tag{53}$$

The longitudinal speed is also constrained to lie within a prescribed range,

$$v_{\min} \;\le\; v_{x,k} \;\le\; v_{\max}, \qquad k = 0, \ldots, N. \tag{54}$$

Optionally, a simple lateral capacity guard derived from tire friction limits is enforced by bounding an approximate lateral force proxy proportional to $v_{x,k}^2 \, \delta_k$:

$$-\,F_{y,\text{cap}} \;\le\; C_k \, \delta_k \;\le\; F_{y,\text{cap}}, \tag{55}$$

where $C_k = m_{\text{total}} v_k^2 / L$ depends on the reference speed and vehicle parameters.

Adaptive Cruise Control (ACC) constraints: When a lead vehicle is present, the spacing state d_k is constrained to remain nonnegative and to satisfy a time-headway rule up to a slack $s_{\text{acc},k}$,

$$d_k \;\ge\; 0, \qquad k = 0, \ldots, N, \tag{56}$$

$$\begin{aligned} d_k - \tau \, v_{x,k} + s_{\text{acc},k} \;\ge\; d_{\min}, \\ s_{\text{acc},k} \;\ge\; 0, \qquad k = 0, \ldots, N-1. \end{aligned} \tag{57}$$

where τ is the desired time headway and $d_{\min}$ is a standstill distance.

TABLE VII: Polynomial coefficients for speed-dependent acceleration bounds.

Function	Coefficient	Value
$a_{\max}(v_x)$	c_0	2.0
	c_1	-0.0425
	c_2	0.000125
	c_3	0.0
$a_{\min}(v_x)$	c_0	-2.5
	c_1	0.0
	c_2	0.0
	c_3	0.0

Acceleration bounds: Finally, speed-dependent longitudinal acceleration limits are imposed through a discrete difference constraint on $v_{x,k}$:

$$a_{\min}\big(v_{x,k}^{\text{ref}}\big) \;\le\; \frac{v_{x,k+1} - v_{x,k}}{\Delta t} \;\le\; a_{\max}\big(v_{x,k}^{\text{ref}}\big), \qquad k = 0, \dots, N-1, \tag{58}$$

where $a_{\min}(\cdot)$ and $a_{\max}(\cdot)$ are polynomial functions of the reference speed from Table VII that encode comfort and powertrain/braking limits.

IV. Results

To evaluate the effectiveness of the proposed MPC framework, three representative driving scenarios were simulated: (i) pure lane following, (ii) lane change with active adaptive cruise control (ACC), and (iii) obstacle avoidance under ACC constraints. These scenarios cover the core functionalities required in highway automation, including steady-state path tracking, smooth lateral maneuvers, and safety-preserving longitudinal regulation. The following subsections summarize the closed-loop behavior and analyze the vehicle dynamics, control inputs, and tire-force responses for each scenario. **For all visualization plots, please refer to the appendix.**

A. Scenario 1: Pure Lane Following (Fig. 8-11 in Appendix)

In the pure lane-following case, the ego vehicle tracks the road centerline without obstacles or ACC intervention. The global trajectory in Fig. 2 shows near-perfect overlap between the reference and actual vehicle paths.

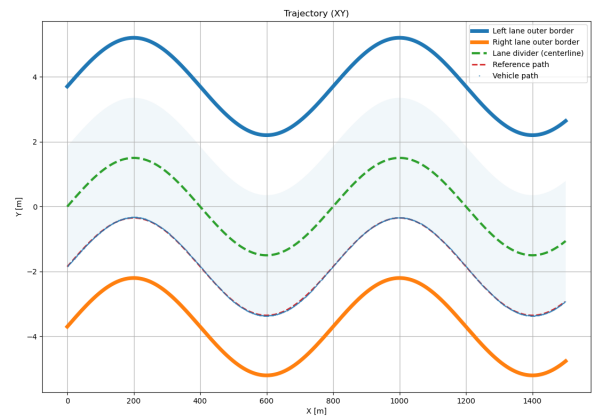
Lateral and heading errors remain extremely small ($|e_y| < 0.03\,\text{m}$, $|e_\psi| < 2\cdot 10^{-3}$ rad), demonstrating high-precision tracking. The vehicle smoothly converges to the desired speed $v_{\text{ref}} = 33\,\text{m/s}$ without overshoot.

Control inputs exhibit steady, stable behavior: the propulsion force decreases from about 900 N to nearly zero as the vehicle reaches steady-state, while steering commands remain very small ($|\dot{\delta}| < 0.02$ rad/s, $|\delta| < 0.008$ rad). Longitudinal and lateral accelerations stay within 1.2 m/s^2.

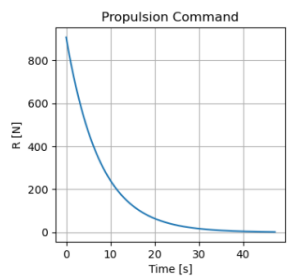
Tire forces remain in a low-slip regime ($|F_y| < 60\,\text{N}$), normal loads vary slightly with transient acceleration, and yaw moments remain modest ($|M_z| < 40\,\text{Nm}$). Analysis of F_y–slip angle, normalized forces, and friction circles confirms that the vehicle operates far from the friction limits, with near-neutral understeer characteristics.

B. Scenario 2: Lane Change With ACC (Fig. 12-15 in Appendix)

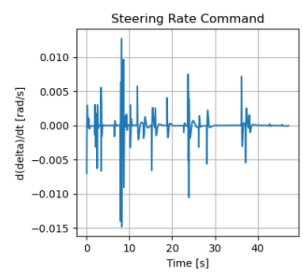
In this scenario, the ego vehicle performs a lane change at $t_{\text{change}} = 10$ s within $T_{\text{change}} = 5$ s while adjusting its speed relative to a slower lead vehicle. Fig. 3 shows a smooth lateral transition between lanes using the C^1 blend, with the vehicle accurately tracking the reference.



(a) Ego vehicle path



(b) Propulsion/Braking force command R_{cmd}



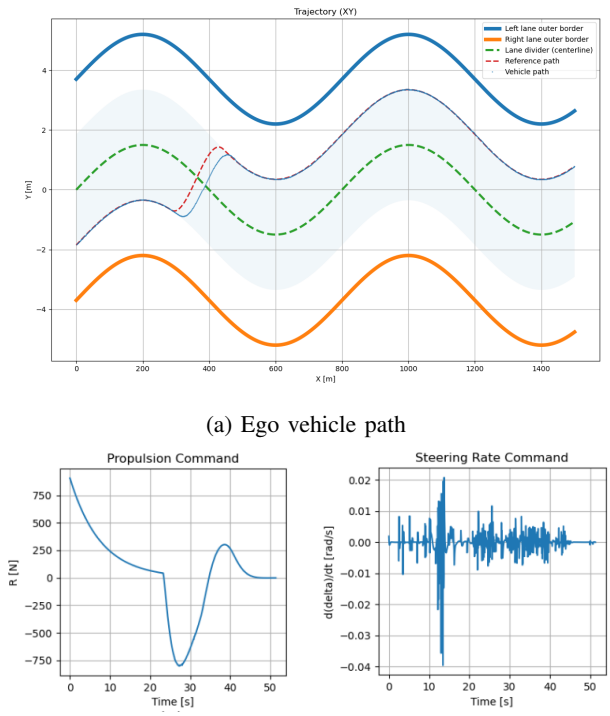
(c) Steering rate command $\dot{\delta}_{\text{cmd}}$

Fig. 2: Ego vehicle path and control commands while following a curvy lane in a free-traffic scenario (Scenario 1.)

During the initial maneuver, the lateral error dips to $e_y \approx -0.9\,\text{m}$, and the heading error reaches $e_\psi \approx 0.035$ rad, as expected from the lane change maneuver. Both return near zero after the transition. Speed tracking illustrates ACC activation: the ego vehicle decelerates around $t \approx 25$ s as the gap becomes closer and stabilizes at the lead vehicle's speed.

Propulsion force shows a strong braking phase ($R \approx -750\,\text{N}$) followed by moderate reacceleration. Lateral acceleration increases during the lane change but remains within $\pm 0.5\,\text{m/s}^2$. Steering rate peaks at about 0.04 rad/s, and steering angle reaches roughly 0.002 rad.

Lateral tire forces grow to about $\pm 300\,\text{N}$, and normal-load shifts reflect combined braking and steering. Yaw moment increases to roughly $\pm 200\,\text{Nm}$. Normalized tire forces remain within friction limits, and the understeer indicator widens during the maneuver but retains stable behavior.

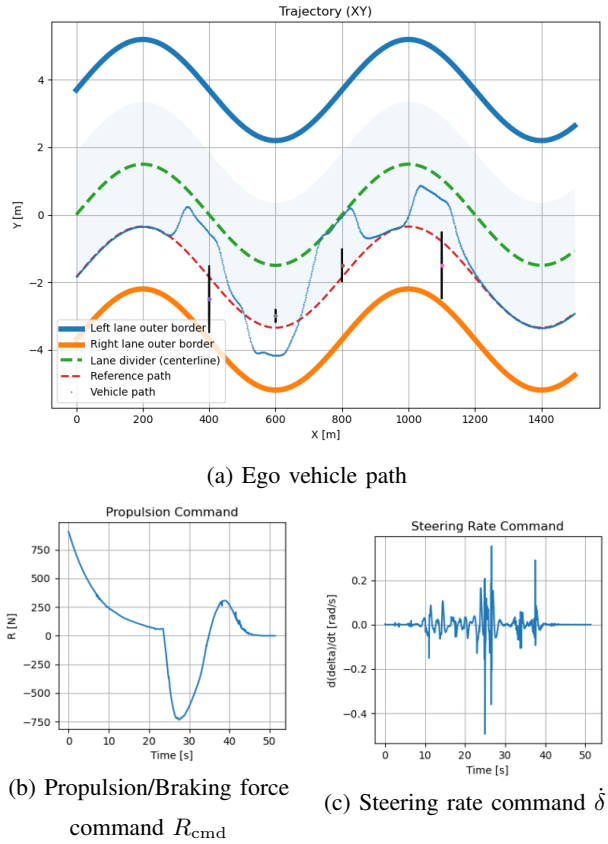


(a) Ego vehicle path

(b) Propulsion/Braking force command R_{cmd}

(c) Steering rate command $\dot{\delta}_{\mathrm{cmd}}$

Fig. 3: Ego vehicle path and control commands while performing a lane change with a lead vehicle in front of the ego vehicle (Scenario 2.)



(a) Ego vehicle path

(b) Propulsion/Braking force command R_{cmd}

(c) Steering rate command $\dot{\delta}$

Fig. 4: Ego vehicle path and control commands while performing an obstacle avoidance with a lead vehicle in front of the ego vehicle (Scenario 3.)

C. Scenario 3: Obstacle Avoidance With ACC (Fig. 16-19 in Appendix)

The third scenario introduces multiple static obstacles and a slower lead vehicle. The ego vehicle must follow a corridor that shifts laterally around the obstacles while maintaining ACC constraints. Fig. 4 shows significant lateral deviations corresponding to corridor geometry.

Lateral error reaches up to $\pm 2\,\mathrm{m}$ during avoidance, and the heading error peaks around $0.05\,\mathrm{rad}$, as expected from the obstacle-avoidance maneuver. Speed ramps up initially and then decreases due to ACC activation ($t \approx 25\,\mathrm{s}$), with further adjustments occurring during obstacle negotiation. The ACC gap remains positive throughout, and the minimum obstacle distance remains above the inflated obstacle radius plus safety margin at all times.

Control inputs become more aggressive: propulsion force reaches $R \approx -900\,\mathrm{N}$ during braking, and lateral acceleration peaks at $1.5\,\mathrm{m/s^2}$. Steering-rate commands rise to $0.1\,\mathrm{rad/s}$. Yaw acceleration and yaw moment show high-frequency variations, with M_z reaching magnitudes over $600\,\mathrm{Nm}$. The peak longitudinal jerk remains within a magnitude of $1.2\,\mathrm{m/s^3}$, which lies below commonly accepted comfort thresholds for seated passengers. Empirical studies on automated-vehicle motion comfort suggest that jerk levels below approximately $1\,\mathrm{m/s^3}$ are perceived as comfortable [18], [19]. Recent test-track experiments further confirm that smoother deceleration profiles with limited jerk are preferred by passengers [20]. Consequently, the observed jerk in our simulation can be considered acceptable from a comfort standpoint.

Tire forces reach the highest levels across all scenarios, with peaks exceeding $\pm 1000\,\mathrm{N}$, though still within the friction ellipse. Slip angles span a much broader range, and the tire-force scatter plots reveal operation near the nonlinear boundary. Despite these demands, the controller maintains feasibility, safety margins, and closed-loop stability.

D. Tire–Dynamics Interpretation Across All Scenarios

Across all three scenarios (Fig. 11, 15, 19), the lateral tire forces exhibit an approximately linear relationship with slip angle. This is expected because the operating conditions remain within the small-slip region of the Magic Formula, where the cornering stiffness dominates and $F_y \approx C_\alpha \alpha$ holds with high accuracy. Even in the

more aggressive Scenario 3, the slip angles remain small enough for the linear approximation to remain valid.

The scatter plots of yaw moment versus lateral acceleration (M_z–a_y) and the understeer indicator $(\alpha_f - \alpha_r)$ display vortex-shaped loops, indicating a mild hysteresis effect. This behavior is characteristic of transient vehicle dynamics: variations in load transfer, steering rate, and time-delayed lateral-force buildup cause the front and rear slip angles to respond at slightly different [21], [22], [23], [24]. As a result, M_z does not follow a purely single-valued function of a_y, and the front–rear slip-angle balance traces a loop rather than a line. Such hysteresis patterns are commonly observed in closed-loop simulations and experimental data and are consistent with physically realistic lateral weight transfer and relaxation effects.

Load-sensitivity plots further confirm correct tire behavior. As the normal load F_z increases, the lateral force capability grows sublinearly due to the load-dependent cornering stiffness of pneumatic tires. This trend appears clearly in all scenarios, particularly during braking and obstacle-avoidance maneuvers where significant longitudinal weight transfer occurs.

Finally, all force samples lie strictly within the friction-ellipse boundaries for both axles, indicating that the tires remain in the adhesion region throughout the simulations. Neither axle approaches saturation, even in Scenario 3, demonstrating that the MPC maintains feasible and physically realistic force demands. This reinforces that the proposed controller respects tire-force limits and preserves stability across a wide range of maneuvers.

V. Discussion

A. Modeling Assumptions

Several simplifying assumptions were adopted in this work to ensure real-time feasibility of the MPC formulation and to maintain a tractable prediction model. Although common in control-oriented vehicle modeling, each assumption carries implications for accuracy, robustness, and applicability. A summary of the key assumptions and their importance is provided below.

a) Perfect state estimation.: All states $(e_y, e_\psi, v_x, v_y, \dot{\psi}, \delta, d)$ are assumed to be perfectly known without sensor noise or estimation delay. This allows the MPC to operate with accurate initial conditions. In practice, IMU drift, GNSS errors, and perception latency introduce uncertainty; neglecting these effects would cause the controller to be overly optimistic and may produce aggressive actions that are sensitive to estimation errors. Robustness or observer design would be required in a real deployment.

b) Dynamic bicycle model with static Magic Formula tires.: The prediction model uses a single-track representation and a quasi-static Magic Formula for lateral forces. This captures steady-state cornering but ignores tire relaxation length and transient force buildup. If this assumption is relaxed, the model would better capture rapid steering maneuvers—the kind that produce hysteresis in F_y–α—and would improve prediction accuracy in high-speed obstacle-avoidance or emergency scenarios.

c) Infinite longitudinal tire stiffness.: The longitudinal slip dynamics are neglected by assuming an instantaneous mapping between propulsion/brake force and longitudinal tire force. This assumption simplifies the model and avoids introducing additional slip states. If violated, the controller may fail to predict transient wheel–ground interactions, especially under hard braking or rapid throttle changes, leading to underestimated stopping distances or incorrect yaw–longitudinal coupling.

d) Neglecting lateral dynamic load transfer.: The normal loads are approximated without roll dynamics or lateral load transfer. This simplifies computation but underestimates peak loads on the outer tires during strong cornering. If lateral load transfer were included, the effective cornering stiffness would vary more significantly, altering stability margins and friction utilization predictions.

e) Fixed longitudinal load distribution (RWD 100:0 under acceleration, 40:60 under braking).: A simplified axle force distribution avoids solving a load-transfer or longitudinal slip model. Violating this assumption—e.g., through aggressive acceleration or uneven surface conditions—would modify available friction at each axle and change the attainable yaw moment. A more complete model would adapt thedistribution dynamically using suspension geometry and pitch dynamics.

f) No system latency.: Both actuation delay and computation delay are ignored. In practice, steering systems, brake actuators, and perception pipelines introduce 50–150 ms of latency. Neglecting latency can lead to instability or oscillatory behavior, and real implementations require delay compensation, previewing, or buffering.

g) Additional implicit assumptions.: Several common assumptions also play a role: (i) a locally planar road geometry, (ii) constant friction coefficient μ, (iii) negligible aerodynamic drag at the speeds considered, and (iv) perfect corridor and obstacle information. Violations of these assumptions could introduce curvature mismatch, force-scaling errors, or incorrect feasibility regions for the MPC.

Overall, these assumptions enable a computationally efficient prediction model suitable for real-time MPC.

However, relaxing any of them would require additional modeling complexity or estimator design, and their impact must be considered when transferring the approach to real vehicles or high-speed operations.

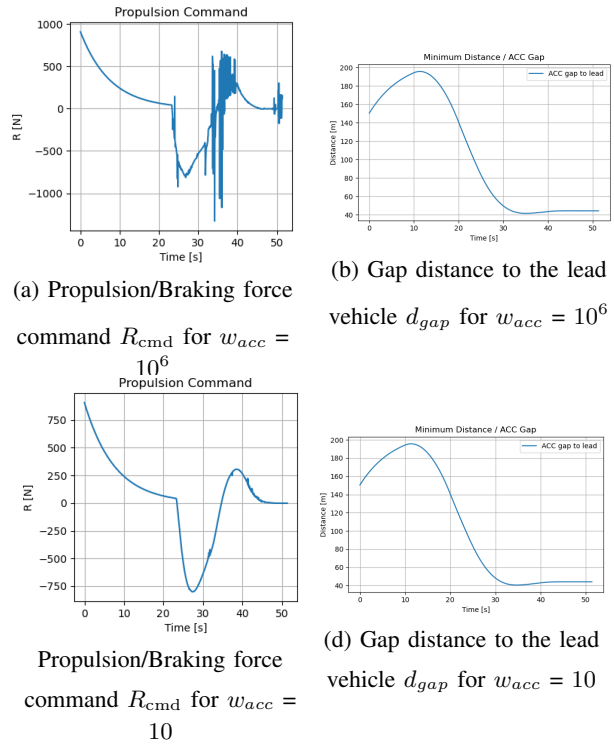
B. Insights From the Case Studies

The three scenarios also provided several practical insights on how modeling, constraints, and tuning choices shape the closed-loop behavior.

a) Effect of ACC slack weighting on propulsion smoothness.: The slack weight w_{acc} plays a critical role in shaping the longitudinal behavior of the controller as shown in Fig. 5. A large value (e.g., $w_{\text{acc}} = 10^6$) effectively hardens the ACC constraint, making any predicted gap violation extremely costly. As a result, when the optimizer anticipates that the ego vehicle may encroach on the desired time gap, it generates an aggressive reduction in the propulsion command R_k to avoid slack usage. Once the gap opens again, the optimizer correspondingly increases R_k to recover speed. This repeated tightening and loosening of the gap induces pronounced step-to-step fluctuations in R_k, as observed in the simulation.

In contrast, a more moderate value (e.g., $w_{\text{acc}} = 10$) allows the controller to tolerate small violations of the nominal gap when necessary, reducing the incentive to oscillate R_k and yielding a significantly smoother longitudinal profile. This illustrates a fundamental safety–comfort trade-off: a very stiff slack penalty enforces stringent adherence to the ACC constraint at the cost of choppy actuation, while a softer penalty improves ride comfort and input smoothness but permits limited gap relaxation. Appropriate tuning of w_{acc} is therefore essential for balancing safety requirements against propulsion smoothness in practical implementations.

b) Effect of corridor smoothing slope.: From Algorithm 1, the corridor smoothing parameter `slope` strongly influences the shape of the obstacle-avoidance path. Smaller values produce smoother, more gradual lateral bounds, which in turn yield a smoother $e_{y,\text{ref}}(s)$ during avoidance but slightly reduce the available corridor width. Larger values preserve sharper geometric features of the raw bounds, enabling tighter maneuvers at the cost of increased curvature and potential oscillations in the planned path. In combination with the PWAref, we observed that a relatively small `slope` significantly reduces oscillations in e_y during obstacle avoidance while still leaving sufficient clearance around the inflated obstacles, as shown in Fig. 6. This parameter needs to be tuned with caution sinch too little `slope` can make the corridor planner unable to find a feasible path to avoid the obstacles.



(a) Propulsion/Braking force command R_{cmd} for w_{acc} = 10^6

(b) Gap distance to the lead vehicle d_{gap} for w_{acc} = 10^6

Propulsion/Braking force command R_{cmd} for w_{acc} = 10

(d) Gap distance to the lead vehicle d_{gap} for w_{acc} = 10

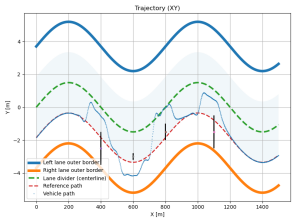
Fig. 5: A comparison of propulsion/braking force command R_{cmd} for a lead vehicle following scenario when w_{acc} = 10^6 and 10.

c) Obstacle avoidance and reference overshoot.: In early tests, the geometric corridor planner alone (Algorithm 1) produced slight overshoot when the vehicle reconciled back to the main path after passing an obstacle. The lateral reference would snap back too aggressively toward the centerline, causing transient increases in e_y and e_ψ. Introducing the piecewise-affine reference adaptation (Algorithm 2), which blends the current lateral error $\hat{e}_{y,k}$ toward the road-based reference over the prediction horizon, largely eliminated this issue. The result in Fig. 7 is a visibly smoother reintegration into the lane with reduced peaks in lateral acceleration and yaw moment.

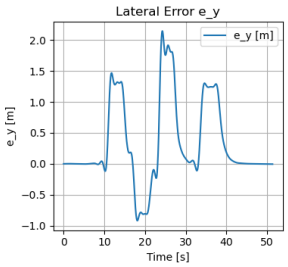
C. Limitations and Directions for Improvement

While the simulation results demonstrate promising performance, several limitations remain before the proposed framework can be deployed in more realistic settings.

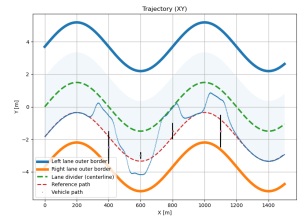
a) Measurement noise and state estimation.: The current study assumes perfect state feedback without sensor noise or latency. In practice, the vehicle state must be estimated from noisy IMU, GNSS, odometry, and perception measurements, often with nontrivial delays. A natural extension is to integrate a



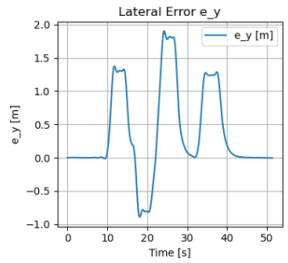
(a) Trajectory plot for `slope` = 0.5



(b) Lateral error plot for `slope` = 0.5

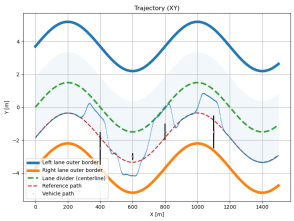


(c) Trajectory plot for `slope` = 0.05

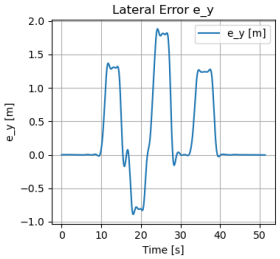


(d) Lateral error plot for `slope` = 0.05

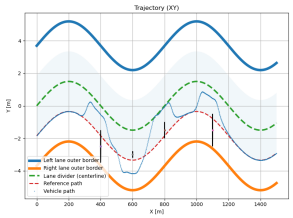
Fig. 6: A comparison of path smoothness while performing an obstacle avoidance before and after for `slope` = 0.5 and 0.05.



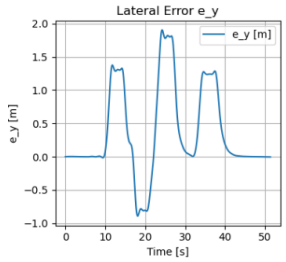
(a) Trajectory plot before applying the `PWAref` algorithm



(b) Lateral error plot before applying the `PWAref` algorithm



(c) Trajectory plot after applying the `PWAref` algorithm



(d) Lateral error plot after applying the `PWAref` algorithm

Fig. 7: A comparison of the reduction of overshoot when returning to the reference after avoiding obstacles by applying Algorithm 2.

dedicated estimator—e.g., an EKF/UKF or a multi-sensor fusion framework—to provide filtered estimates of $(e_y, e_\psi, v_x, v_y, \dot{\psi}, \delta, d)$ and associated covariance information. This would also enable robust or stochastic MPC formulations that explicitly account for uncertainty in the predicted states and tighten constraints accordingly.

b) Richer vehicle and tire models.: The prediction model is limited to a planar dynamic bicycle with quasi-static Magic Formula tires and simplified load transfer. More complex models could capture additional phenomena such as roll and pitch dynamics, lateral load transfer, combined-slip effects with relaxation length, and aerodynamic forces at higher speeds. Incorporating these effects would improve prediction accuracy in extreme maneuvers (hard braking, rapid lane changes, low-friction surfaces), at the expense of increased computational complexity and more challenging linearization.

c) More complex environments and traffic scenarios.: The simulated scenarios involve a single ego vehicle, a single lead vehicle, and static obstacles on a relatively simple road geometry. In realistic highway environments, the controller must handle multi-lane networks, ramps and merges, cut-in/cut-out maneuvers from surrounding traffic, time-varying speed limits, and spatially varying friction (e.g., wet patches, ice, or uneven pavement). Extending the corridor and ACC formulations to multi-vehicle interaction, priority rules, and dynamic right-of-way constraints would be a key step toward deployment in dense traffic.

d) Latency, implementation, and real-time optimization.: System latency and solver computation time are neglected here. Future work should incorporate actuation and perception delays into the prediction model and investigate delay-compensation strategies. On the numerical side, exploiting structure (warm-starting, problem-specific code generation, and tailored QP solvers) is crucial to guarantee real-time feasibility on automotive-grade ECUs. Adaptive tuning of cost weights and constraint tightening based on online performance metrics could further improve robustness and comfort.

e) Learning-based adaptation.: Finally, many parameters in the current framework (tire models, friction coefficients, cost weights) are fixed a priori. Data-driven or learning-based adaptation—e.g., online friction estimation, driver- or fleet-specific preference learning, or reinforcement learning for high-level parameter scheduling—could allow the controller to adjust to changing conditions and individual comfort/safety preferences while retaining the structure and guarantees of MPC at the lower level.

VI. Conclusion

This study presented a model predictive control framework capable of addressing the coupled longitudinal–lateral demands of highway driving, including lane following, lane changing, adaptive cruise control, and obstacle avoidance. The formulation integrates a dynamic bicycle model with Magic-Formula-based tire forces, a friction-ellipse constraint for combined slip, and a geometric corridor-planning method that ensures feasibility with respect to road boundaries and obstacles.

Simulation results across three representative scenarios demonstrate the effectiveness of the approach. In pure lane following, the controller achieves extremely high lateral accuracy and smooth longitudinal regulation. During lane changes with ACC, the framework maintains safe headway while tracking a blended lane reference with stable yaw and tire-force behavior. In the most demanding scenario with obstacle avoidance under ACC constraints, the controller successfully negotiates narrow corridors and maintains feasibility despite significant load transfer, increased lateral forces, and aggressive steering inputs. Across all scenarios, tire forces remain inside the friction ellipse, and the vehicle stays within a physically consistent and dynamically stable operating regime.

The insights gained from the study highlight the importance of appropriate slack-weight tuning for balancing safety and comfort, the need for smooth corridor construction during avoidance, and the influence of the PWA reference on the improvement of transitions between avoidance maneuvers and nominal lane tracking. Although the current framework relies on several simplifying assumptions, it provides a strong foundation for future extensions involving uncertainty, richer vehicle models, and more complex traffic environments.

Overall, the results indicate that MPC, combined with geometric corridor planning and structured constraint handling, offers a powerful and generalizable tool for high-performance autonomous driving on structured roadways.

References

[1] American Association of State Highway and Transportation Officials, *A Policy on Geometric Design of Highways and Streets*. Washington, D.C.: AASHTO, 2018.

[2] Federal Highway Administration, "Roadway design manual: Geometric design standards," https://www.fhwa.dot.gov, 2020.

[3] M. Nezami, H. S. Abbas, N. T. Nguyen, and G. Schildbach, "Robust tube-based lpv-mpc for autonomous lane keeping," *IFAC Journal of Systems and Control*, vol. 22, pp. 100–121, 2022.

[4] J. Dallas, J. Wurts, J. L. Stein, and T. Ersal, "Contingent nonlinear model predictive control for collision imminent steering in uncertain environments," in *Proceedings of the ASME Dynamic Systems and Control Conference (DSCC)*. Park City, UT: ASME, 2019.

[5] J. Liu, P. Jayakumar, J. L. Stein, and T. Ersal, "Combined speed and steering control in high-speed autonomous ground vehicles for obstacle avoidance using model predictive control," in *Proceedings of the American Control Conference (ACC)*, 2016, pp. 2553–2558.

[6] H. B. Pacejka, *Tire and Vehicle Dynamics*, 3rd ed. Oxford, UK: Butterworth-Heinemann, 2012.

[7] G. N. Dell'Annunziata *et al.*, "A new approach for estimating tire–road longitudinal forces," *Vehicle System Dynamics*, 2019, models a rear-wheel-drive vehicle with traction applied on the driven axle.

[8] K. Bayar *et al.*, "Modelling and control of a brake system for an extended-range electric vehicle," *Control Engineering Practice*, 2012, compares against a 60/40 front/rear parallel braking strategy.

[9] T. D. Gillespie, *Fundamentals of Vehicle Dynamics*. SAE International, 1992.

[10] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. Springer, 2012.

[11] P. Falcone, F. Borrelli, J. Asgari, H. Tseng, and D. Hrovat, "A mpc-based vehicle handling controller for stability enhancement," in *46th IEEE Conference on Decision and Control*, 2007.

[12] T. Toledo, D. Zelniker, A. Polus, and R. Livneh, "Modeling duration of lane changes," *Transportation Research Record*, no. 1999, pp. 71–78, 2007.

[13] M. G. Plessen, D. Bernardini, H. Esen, and A. Bemporad, "Spatial-based predictive control and geometric corridor planning for adaptive cruise control coupled with obstacle avoidance," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 1, pp. 38–50, 2018.

[14] R. Milton and P. Varaiya, "Adaptive cruise control: System overview, design issues and future trends," *IEEE Transactions on Intelligent Transportation Systems*, 2017.

[15] G. J. L. Naus, J. Ploeg, M. J. G. van de Molengraft, and M. Steinbuch, "String-stable adaptive cruise control: Design, analysis and experiments," in *IEEE American Control Conference*, 2010.

[16] G. H. Bham and R. F. Benekohal, "Analysis of time headways in traffic flow: Empirical studies and implications for acc design," *Transportation Research Record*, 2010.

[17] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Physical Review E*, 2000.

[18] K. N. de Winkel, T. Irmak, R. Happee, and B. Shyrokau, "Standards for passenger comfort in automated vehicles: Acceleration and jerk," *Applied Ergonomics*, vol. 106, p. 103881, 2023.

[19] I. Bae *et al.*, "Toward a comfortable driving experience for a self-driving shuttle," *Electronics*, vol. 8, no. 9, p. 943, 2019.

[20] S. Carlowitz *et al.*, "User evaluation of comfortable deceleration profiles for highly automated driving," *Transportation Research Part F: Traffic Psychology and Behaviour*, 2024, open access.

[21] I. Kobayashi, J. Kuroda, D. Uchino, K. Ogawa, K. Ikeda, T. Kato, A. Endo, M. H. B. Peeie, T. Narita, and H. Kato, "Research on yaw moment control system for race cars using drive and brake torques," *Vehicles*, vol. 5, no. 2, pp. 515–534, Apr. 2023.

[22] K. Hulme, E. Kasprzak, K. English, D. Moore-Russo, and K. Lewis, "Using gaming and motion simulation to enhance vehicle dynamics education," in *ASEE Annual Conference & Exposition*, 2009.

[23] M. Yao, D. Lu, H. Yin, L. Li, M. Lv, and W. Wang, "The unsteady-state response of tires to slip angle and vertical load variations," *Machines*, vol. 10, no. 7, p. 527, 2022.

[24] R. Guo, Z. Jiang, and L. Yuan, "Application of steering robot in the test of vehicle dynamic characteristics," in *Proceedings of the 3rd International Conference on Mechatronics, Robotics and Automation (ICMRA)*, Tianjin, China, 2015.

APPENDIX
ADDITIONAL SCENARIO VISUALIZATIONS

Scenario 1: Pure lane following on free roads

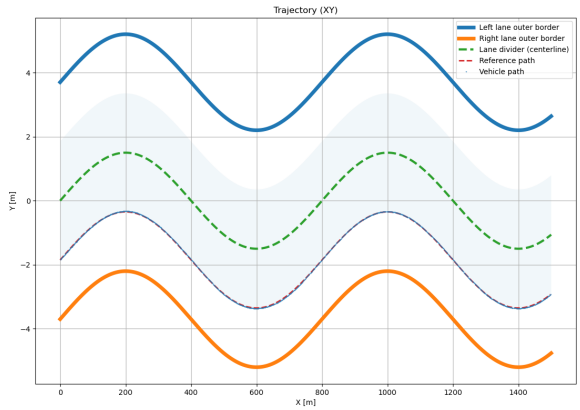


Fig. 8: Trajectory overview for Scenario 1.

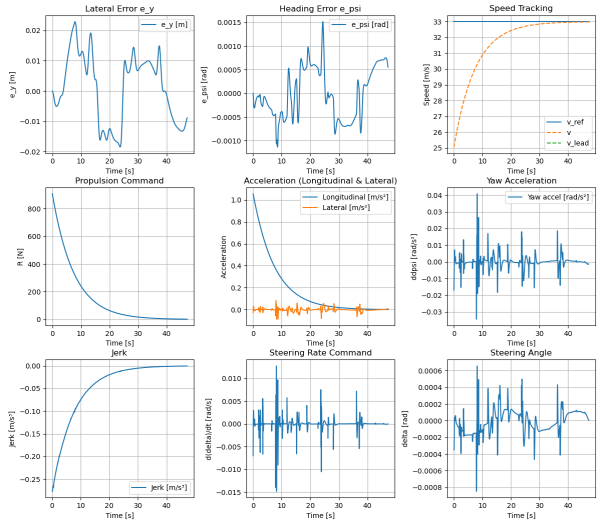


Fig. 9: Vehicle dynamic response variables for Scenario 1.

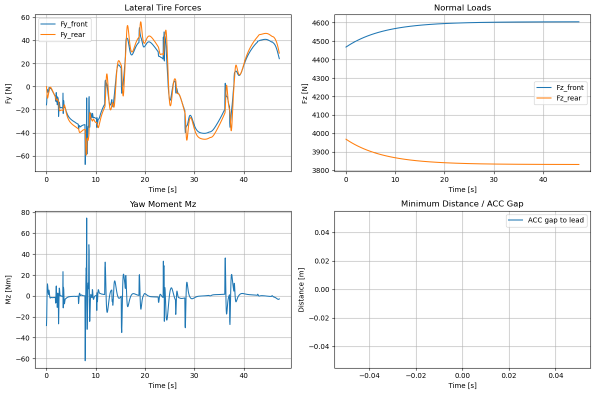


Fig. 10: Tire forces and normal loads for Scenario 1.

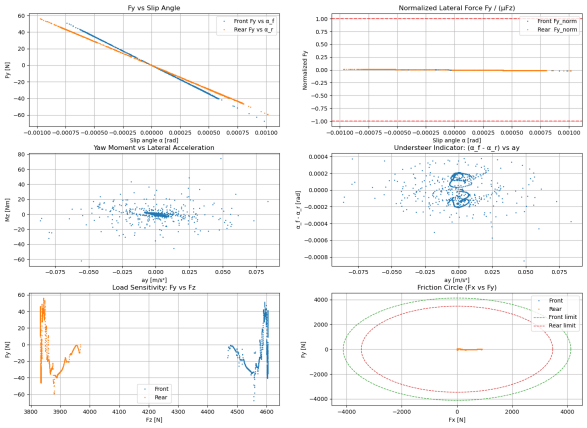


Fig. 11: Friction circle and tire characteristics for Scenario 1.

Scenario 2: Lane change with lead vehicle

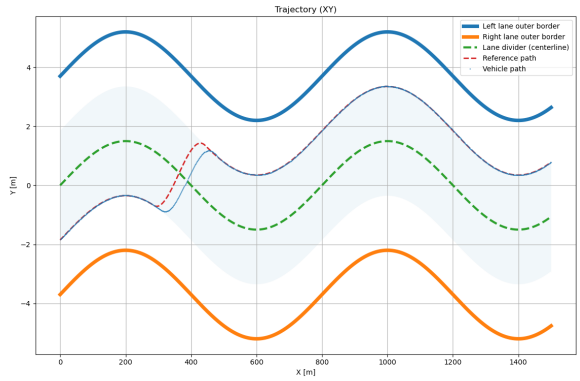


Fig. 12: Trajectory overview for Scenario 2.

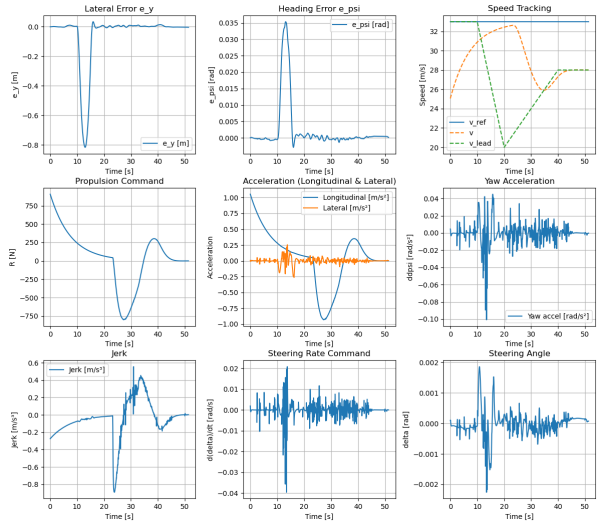


Fig. 13: Vehicle dynamic response variables for Scenario 2.

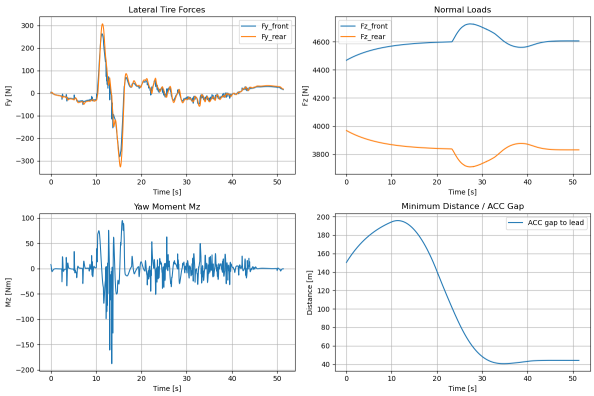


Fig. 14: Tire forces and normal loads for Scenario 2.

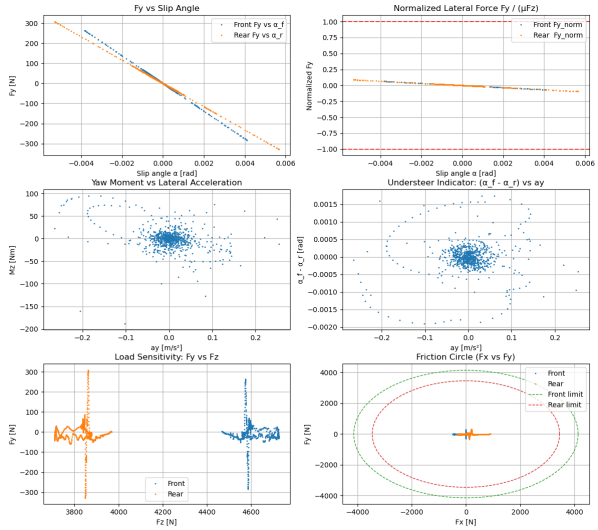


Fig. 15: Friction circle and tire characteristics for Scenario 2.

Scenario 3: Obstacle avoidance with lead vehicle

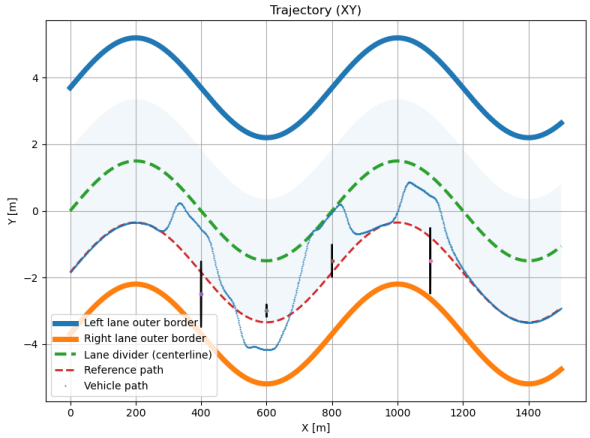


Fig. 16: Trajectory overview for Scenario 3.

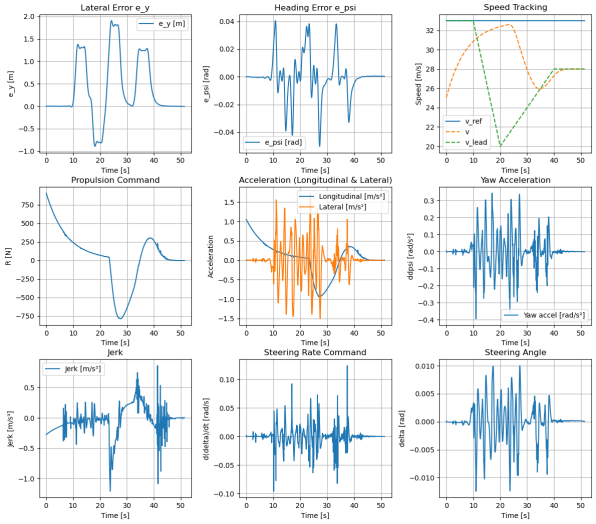


Fig. 17: Vehicle dynamic response variables for Scenario 3.

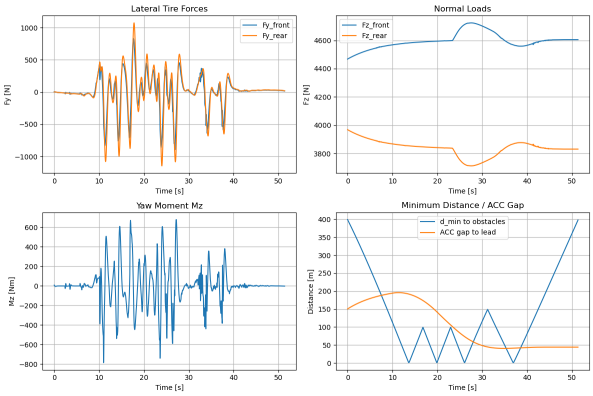


Fig. 18: Tire forces and normal loads for Scenario 3.

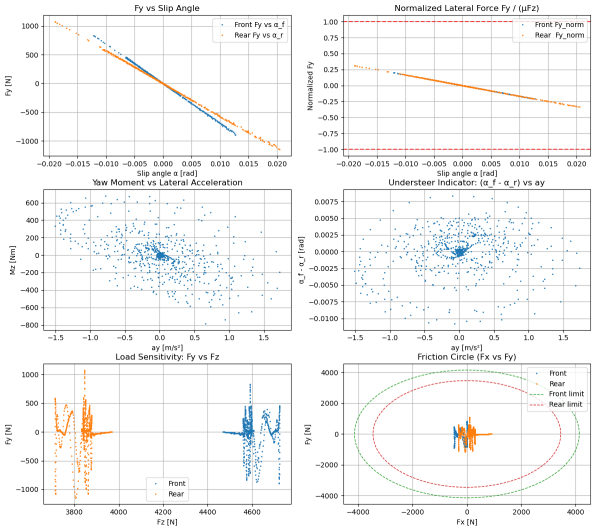


Fig. 19: Friction circle and tire characteristics for Scenario 3.